

Abstract:

A target star light curve analysis through the transit method determines the existence of exoplanets in this study. The periodic brightness dips establish the presence of multiple planets existing in the system. The transit data enables us to calculate orbital parameters, which include period size together with distance. The calculation of surface temperature proceeds based on an assumption of zero greenhouse effect. We analyze the ability of these exoplanets to support life through studying existing evidence about atmospheric elements that dictate their habitability status.

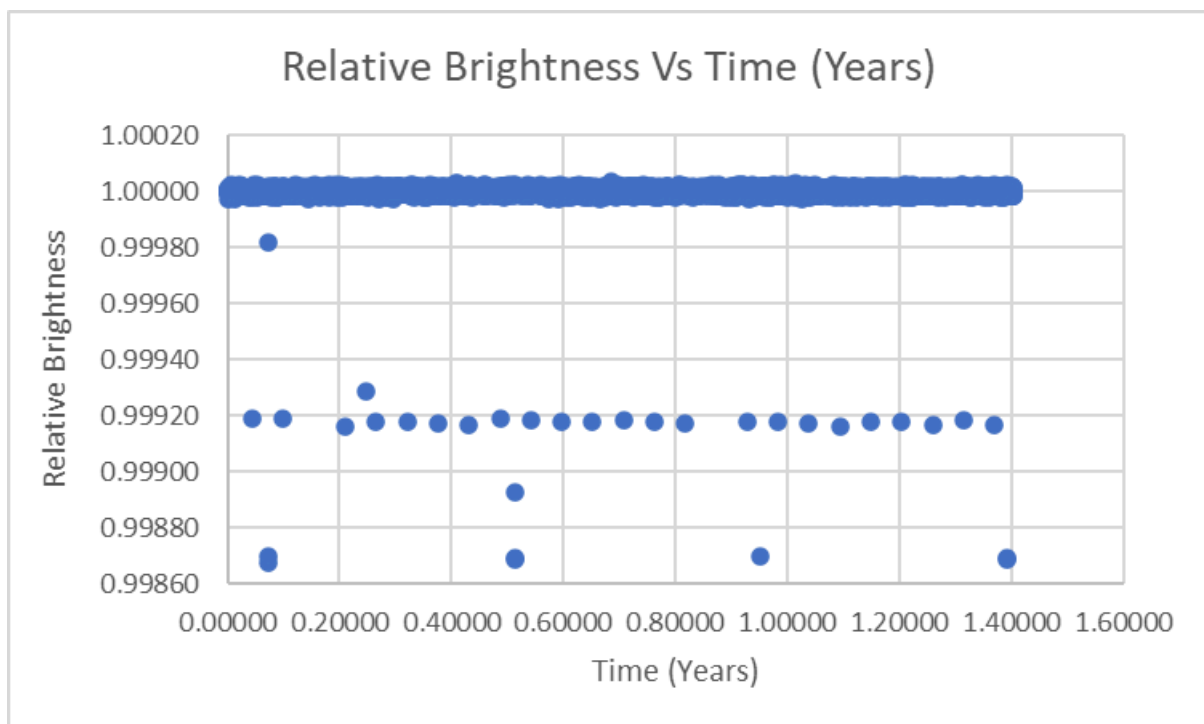
Introduction:

Astronomers and scientists have reached a major point of scientific breakthrough by discovering exoplanets, which lie beyond the planets of our solar system. Sunlight blocking at exoplanets allows astronomers to better understand their formation, besides gaining knowledge about the atmospheres and life conditions existing elsewhere in our universe. Scientists use the transit method as an outstanding tool to search for exoplanets because it measures periodic brightness changes in stars when planets obstruct the star's light. Through this procedure, scientists can successfully gather important details, especially details such as how large the planet is as well as its orbital period and atmospheric characteristics. Thousands of exoplanets discovered through the transit method become visible because of the Kepler missions operated by NASA. Using light curve analysis allows researchers to extract important planetary data, which helps determine habitable conditions. The objective of this investigation is to evaluate the provided light curve data for transit events while determining the life sustaining potential of the detected planets.

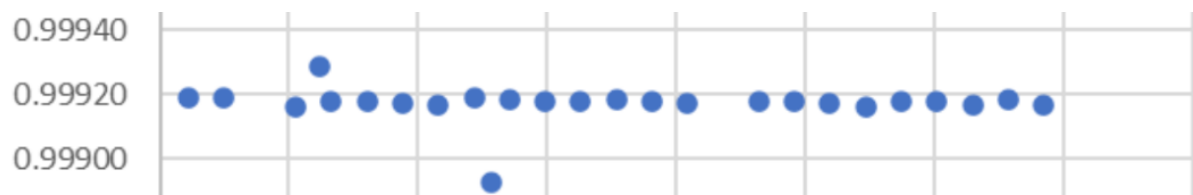
Results:

a) It is established that the target star hosts two exoplanets which orbit around it. The light curve data reveals two distinct brightness patterns that show consistent periodic declines which have different depths and orbital patterns. Regular orbital motions produce the observed repeating dips in the data because they match the signature of planetary transits.

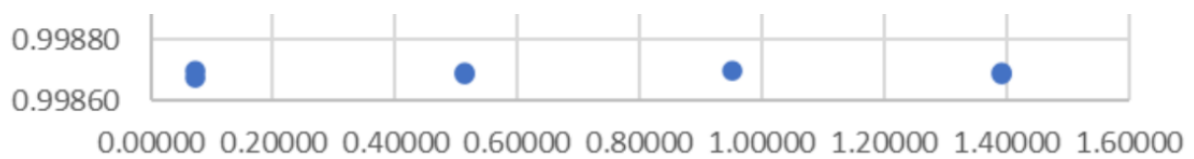
b) The data shows six dips in the graph that are potential exoplanets, which remain unverified since they have recorded unpredictable dips. There is uncertainty about what is causing the light curve those specific dips, but these dips may be exoplanets that need further observational data to be verified. The absence of periodicity confirms and sort of rules out the possibility of those dips being evidence of existing exoplanets.



Exoplanet 1:



Exoplanet 2:



(Part c and d)	Exoplanet 1	Exoplanet 2
Orbital Periods	$P = 0.05194571853$	$P = 0.43879450549$
Transit Depth	Approx. = 0.999192639779523	Approx. = 0.998698165601629
Radius	9690.339513 km	12305.04457 km
Orbital Distance	$1.65242478 \cdot 10^{10} \text{ m}$ OR 0.11045777 AU	$6.85678475 \cdot 10^{10} \text{ m}$ OR 0.45834775 AU
Surface Temperature	197 Celsius	-42 Celsius

Showing Calculations for Exoplanet 2:

Orbital Period:

- The P or orbital period value was calculated by getting the average of the time difference between each dip on the graph. The values were in the CSV file.
- $P = ((4\text{th time} - 3\text{rd time}) + (3\text{rd time} - 2\text{nd time}) + (2\text{nd time} - 1\text{st time})) / 3$
- $P = 0.43879450549$ years

Transit Depth:

- $\text{Transit Depth} = 1 - \text{Relative Brightness During Transit}$
- $\text{Transit Depth} = 1 - 1 - 0.998698165601629 = 0.001301834398$

Radius:

- $R_{\text{star}} = 0.49 * 696\,000 = 341\,040$ km
- Using the formula: $\text{Transit Depth} = R_{\text{planet}}^2 / R_{\text{star}}^2$
- Rearrange to solve for the radius of the planet: $R_{\text{planet}} = \sqrt{\text{Transit Depth}} * R_{\text{star}}$
- $R_{\text{star}} = 0.49 * 696\,000 = 341\,040$ km
- $R_{\text{planet}} = \sqrt{0.001301834398} * 341\,040 = 12305.04457$ km

Orbital Distance:

- Using Kepler's third law: $P^2 = (4\pi^2 * a^3) / (G * M)$

$$P^2 = \frac{4\pi^2}{GM_{\text{star}}} a^3$$

- Rearrange to solve for a: $a = ((G * M * P^2) / (4\pi^2))^{(1/3)}$
- $M = 0.5 * (1.989 * 10^{30}) = 9.945 * 10^{29}$

- P has to be in seconds, so: $P = 0.43879450549 * 365.25 * 24 * 60 * 60 = 13847301.49 \text{ s}$
- $G = 6.674 * 10^{-11}$
- $a = (((6.674 * 10^{-11}) * (9.945 * 10^{29}) * (13847301.49)^2) / (4 * \pi^2))^{(1/3)}$
- $a = 6.85678475 * 10^{10} \text{ m OR } 0.45834775 \text{ AU}$

Surface Temperature:

- Using <https://astro.sitehost.iu.edu/ala/PlanetTemp/index.html>:
- $M = 0.5$
- $A = 0.29$ (Assuming it's similar to Earth)
- $a = 0.45834775 \text{ AU}$
- $T = -42 \text{ Celsius}$

Analysis:

e) Both exoplanets seem to be very high or very low in temperature, but the exoplanet 2 has a chance of Earth-like habitability since its surface temperature is -42 degrees, without factoring in any conditions like a greenhouse effect. On the other hand, exoplanet 1 is extremely high in temperature at 197 degrees, which is much higher than the boiling point of water, making it semi-impossible to have Earth-like life on it.

f) The analysis would be different if the exoplanets had a 90.00 atm that is 96.00% carbon dioxide, because that would highly increase the surface temperature on the plane, which can cause the planet to trap an insane amount of heat and reduce the chance of habitability that could potentially occur or has occurred.

g) Regarding exoplanet 1, confidence is high that it is uninhabitable, since the temperature on the planet is extremely high. But for exoplanet 2, the confidence level is not

very high, because the planet's temperature is not extreme, which makes its temperature not enough to conclude if it is habitable or not, therefore more information such as albedo data or atmospheric composition and pressure for example would help better understand the planet and its chance of habitability.